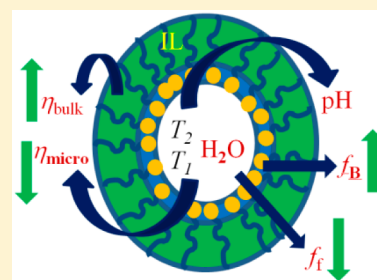


Phase Behavior, Diffusion, Structural Characteristics, and pH of Aqueous Hydrophobic Ionic Liquid Confined Media: Insights into Microviscosity and Microporosity in the $[C_4C_4im][NTf_2]$ + Water System

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ABSTRACT: We present our studies on the physicochemical properties of water confined in Dibutylimidazolium bis(trifluoromethanesulfonylimide) ($[C_4C_4im][NTf_2]$) reverse micelles through the NMR relaxation measurements that provide us an understanding of microviscosity and pH in the confined condition. We present experimental results on phase behavior, diffusion, structural characteristics and pH in aqueous ionic liquid-confined media. The ternary phase diagram was constructed by the cloud point measurements and the microheterogeneous regions were detected by the measurement of bulk viscosity and diffusion coefficients of $K_4[Fe(CN)_6]$ inside the homogeneous microemulsion systems through the cyclic voltammetric (CV) measurements. The size of the microemulsion systems was characterized by the dynamic light scattering (DLS) method. The 1H NMR spectra of homogeneous microemulsion systems were taken which indicates the presence of bound and free water molecules inside the microemulsion system. The NMR spin–lattice relaxation time (T_1) of water molecules in its homogeneous microemulsion systems were measured and the reorientational correlation time (τ_c) of water molecules obtained from it indicates that the fluidity of homogeneous confined media decreases with the decrease in the composition of water. Microviscosity of the aqueous confined media was calculated from the measured T_1 relaxation time values by applying the Debye–Stokes equation and correlated with the bulk viscosity of the samples. It was observed that both the microviscosity and bulk viscosity show inverse relationship. The fraction of bound and free water molecules were calculated from the measured T_1 values. NMR spin–spin relaxation time (T_2) of water molecules in its homogeneous microemulsion systems were measured with the varying pH of the aqueous core. A change in the T_2 relaxation time of the water proton was observed proposing an exchange of proton between the H_2O and $-OH$ group of the TX-100 molecules. Finally, methyl orange (MO) was used as a UV–vis spectrophotometric molecular probe and the measured λ_{max} values of the probe were used for the detection of micropolarity of the homogeneous aqueous confined media and was found to be increase with the increase in the size of the confined media.



INTRODUCTION

Confined water is ubiquitous in nature like in cellular organelles of biological species,¹ water inside reverse micelles,^{2–7} and microemulsions.^{8–14} Water present inside the reverse micelles or microemulsions is of great interest to chemists, biochemists, and biophysicists. Reverse micelles or microemulsions are thermodynamically stable and are isotropic mixture of oil and water with a surfactant or a mixture of surfactants and frequently in combination with a cosurfactant.¹⁵ Structurally there are three different kinds of microemulsions are present; water in oil (W/O) where water is the dispersed phase and oil is the dispersion medium, oil in water (O/W) where oil is the dispersed phase and water is the dispersion medium, and the bicontinuous structure where both the oil and water are present as the continuous phase.^{15–17} Many reports are available in the literature regarding the study of aqueous^{8–14} and nonaqueous¹⁸ microemulsions with oil as a dispersion medium. However, the use of volatile organic solvents for the preparation of microemulsions or reverse micelles is not environmentally benign process. Therefore, according to the green chemistry point of view ionic liquid is used to replace the use of these volatile organic solvents. Ionic liquids have insignificant vapor

pressure, recyclability, high thermal stability, wide electrochemical window, wide liquidus range and nonflammability, to name a few.^{19–23} Ionic liquids are used in the preparation of micelles and microemulsions and several reports are available in the literature for the preparation and characterization of ILs-based microemulsions.^{13,16,17,24–30} Reports are available in the literature for the replacement of water by ILs due to the requirement of high thermally stable microemulsions.^{18,31} In IL-based microemulsions ILs are either used as hydrophobic solvents by the replacement of oil phase^{13,16,17} or used as the hydrophilic phase^{25–30} by replacing the aqueous phase. The most studied hydrophobic IL in case of aqueous ILs-based microemulsions is the $[C_4mim][PF_6]$.^{13,16,17} However, reports are available in the literature regarding the formation of HF molecules by the hydrolysis of anionic part of $[C_4mim][PF_6]$ ^{32,33} which makes the system acidic. This is one of the most fundamental limitation of using $[PF_6]^-$ as anion in ILs for the preparation of microemulsions which may gives the

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incorrect information about the physicochemical properties of aqueous confined media. Therefore, in this report we have replaced the use of this $[\text{PF}_6]^-$ anion-based ILs by choosing a $[\text{NTf}_2]^-$ -based IL with dibutylimidazolium ($[\text{C}_4\text{C}_4\text{im}]^+$) as the cation which are stable toward the hydrolysis. Reports are available in the literature regarding the physicochemical properties of water in hydrophobic IL microemulsions.^{16,34} However, to the best of our knowledge there has not been a single report is available in the literature regarding the physicochemical properties of aqueous microemulsions containing hydrophobic $[\text{NTf}_2]^-$ anion-based ionic liquids. Here, in this report, for the first time we have studied the physicochemical properties of $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ microemulsions, and the outline of the paper is as follows.

First the ternary phase diagram was constructed by taking the hydrophobic $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]$ IL with water in combination with a nonionic surfactant, Triton X-100 to stabilize the biphasic system. Then the microheterogeneous regions were detected by the measurement of diffusion coefficients of an electrochemically active probe through the cyclic voltammetric measurements. Dynamic light scattering technique (DLS) was used for the study of structural features of microemulsions. NMR relaxations measurements were performed to understand the microviscosity and pH of the aqueous IL confined media. Finally, an UV-vis spectrophotometric probe was used to study the micropolarity of the aqueous confined media.

EXPERIMENTAL SECTION

Materials. Materials required for the synthesis of ionic liquid like 1-butyl imidazole (99.9% purity), butyl bromide (99.9% purity) and LiNTf_2 (99.98% purity) were purchased from Sigma-Aldrich. Nonionic surfactant, Triton X-100 (98.5% purity) was purchased from Fluka Analyticals. $\text{K}_4\text{Fe}(\text{CN})_6$ (99.95% purity), used as an electrochemically active probe for the cyclic voltammetric study was purchased from Sigma-Aldrich. Methyl orange (MO) (99% purity) used for the UV-vis spectrophotometric study was purchased from Sigma-Aldrich. Distilled deionized water having electrical resistivity of $18.2 \text{ M}\Omega \text{ cm}$ was obtained from a Millipore Milli-Q unit for the preparation of microemulsions.

Synthesis of Ionic Liquid. The hydrophobic dibutylimidazolium bis(trifluoromethanesulfonyl)imide ($[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]$) ionic liquid was synthesized according to the reported procedure.³⁵ First the quaternization reaction between the butylimidazole and butyl bromide yields the dibutylimidazolium bromide $[\text{C}_4\text{C}_4\text{im}]\text{Br}$. In the next step the exchange of anion between the Br^- of $[\text{C}_4\text{C}_4\text{im}]\text{Br}$ and NTf_2^- of LiNTf_2 (metathesis reaction) takes place in water as the solvent medium. Then the prepared IL was separated through a separating funnel with dichloromethane (DCM). High vacuum was used to remove the excess amount of solvent (both the water and DCM) from the IL. It was ensured that the water content in each IL sample was less than 50 ppm. ^1H NMR spectra was taken for the synthesized ionic liquid which did not show any trace amount of impurity in the sample. The IL was also checked for any halide impurities which was less than 50 ppm.³⁶

Preparation of Microemulsions. $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ microemulsions were prepared by fixing $I = 0.05$ and 0.1 ($I = [\text{IL}]/[\text{TX-100}]$, weight ratio) with the varying weight percent of water. Then the solutions were stirred in a vortex mixer up to the formation of clear, homogeneous and optically transparent solution. Those solutions were kept over months

and no phase separation was observed which indicates the stability of the prepared microemulsions.

APPARATUS AND PROCEDURE

Phase Equilibria. Solubility measurements were performed to construct the ternary phase diagram for $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$. The samples were placed in a glass jacket with an inlet–outlet water circulating system connected externally to a Julabo thermostat to regulate the temperature of the system. All the measurements were performed at a temperature of $298.15 \pm 0.01 \text{ K}$. The ternary phase diagram for the $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ system was constructed through the visual, cloud-point method of solution by observing the change in the cloud point of the solutions.³⁷

Cyclic Voltammetry. Electrochemical properties were investigated by cyclic voltammetric (CV) technique¹⁷ using a BioLogic Science Instrument in a conventional three-electrode cell, where Pt foil was used as the counter electrode, Ag/AgCl as the reference electrode and glassy carbon as the working electrode (WE). Before its use (WE) in the electrochemical study it was polished with a $0.3 \mu\text{m}$ aluminum oxide slurries and was sonicated several times alternately with deionized water and ethanol and dried under lamp. The area of the WE was determined using cyclic voltammetric experiment by taking a reversible system of $4 \text{ mM } \text{K}_3\text{Fe}(\text{CN})_6$ in 1 M KCl solution with the diffusion coefficient $D = 6.36 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$.¹⁷ The calculated surface area of the working electrode was found to be $1.53 \times 10^{-2} \text{ m}^2$. The potential was scanned between $+0.5$ to -0.5 V and the sweep rate were changed from 25 to 100 mV s^{-1} . Potassium ferrocyanide $\text{K}_4\text{Fe}(\text{CN})_6$ was used as the electrochemically active probe in all the microemulsion systems and the temperature used throughout the study was kept constant as $298.15 \pm 0.1 \text{ K}$.

Dynamic Light Scattering. The hydrodynamic diameters of aqueous ionic liquid microemulsions were determined by dynamic light scattering technique (BIC 90 Plus Particle Size Analyzer, Brookhaven Instruments Corporation, USA). All the microemulsion samples were filtered through the $0.22 \mu\text{m}$ PTFE membrane filters procured from RanDisc. The samples were prepared under the condition of inert atmosphere and were taken in a cuvette. Precautions were made to avoid any kind of bubbles formed inside the microemulsion samples. All the measurements were made with a scattering angle of 90° at a temperature of $298.15 \pm 0.1 \text{ K}$.

UV-Vis Spectroscopy. The stock solutions of the methyl orange (UV-vis spectroscopic probe) were prepared in the methanol prior to their use. The stock solution was added dropwise into the microemulsion samples and the methanol (solvent) was evaporated under vacuum. The sample containing the probe was transferred to a quartz cuvette under a nitrogen atmosphere and sealed with a septum. The λ_{max} values were measured at 298.15 K by using a UV-visible spectrophotometer, and the temperature of the cell was monitored using a single cell accessory with an accuracy of $\pm 0.01 \text{ K}$.

pH Measurements. Different pH solutions were prepared by titrating 0.1 M NaOH (98.5% purity, Fluka) with $0.5 \text{ M H}_2\text{SO}_4$ (98.5% purity, Fluka). A digital pH meter was used (Global Digital pH Meter, Model No-504) to measure the pH of the aqueous core (pH of the aqueous solutions were measured externally prior to the formation of microemulsions) in the range of 2.0 – 9.5 . Buffer solutions were used for the calibration of the pH meter. Two buffer solutions (Sigma-

Aldrich) were selected for this purpose; one is an alkaline buffer solution ($\text{pH } 10.3 \pm 0.01$ at 25°C) and the other one is an acidic buffer ($\text{pH } 4.00 \pm 0.01$ at 25°C). All the pH data points are the average of triplicate measurements with a precision of ± 0.01 pH units.

Viscosity Measurements. A Brookfield-ultra rheometer with cone plate arrangement was used for the measurement of viscosity of aqueous ionic liquid microemulsion systems. Temperatures of the samples were monitored by a Julabo constant temperature thermostat bath with an accuracy of ± 0.01 K. The viscosity (η) values of the solutions were obtained through the equation as

$$\eta = (100/\text{RPM})(\text{TK})(\text{torque})(\text{SMC}) \quad (1)$$

where RPM, TK (0.09373), and SMC (0.327) are the speed, viscometer torque constant, and spindle multiplier constant, respectively. The calibration of the instrument was performed by taking the reported viscosity data of aqueous solutions of CaCl_2 , MgCl_2 , and $[\text{C}_4\text{mim}][\text{BF}_4]$ with different concentration.^{38,39} The accuracy of the instrument was obtained as $\pm 1\%$.

NMR Study. All NMR experiments were performed on a Bruker-400 NMR spectrometer operating at a ^1H resonance frequency of 400 MHz, at 298.15 ± 0.1 K. Analysis of neat samples were performed in a 5 mm (outside diameter) NMR tube using D_2O in a capillary tube as external lock. All the spectra were collected by using a one pulse sequence, $[30^\circ\text{-flip angle}]$ with 32k complex data points. A spectral width of 20 ppm and a repetition time of 2 s were used with the averaging of eight signals. Heteronuclear single quantum coherence (HSQC) experiment was performed to differentiate the water signal from other signals present in the microemulsion system. Spin–lattice relaxation times (T_1) were measured by inversion recovery method, $[180^\circ_x - \tau - 90^\circ_x - \text{acq}]$, performing 34 experiments with delay time (τ) ranging from 2 ms to 5 s. Spin–spin relaxation times (T_2) were measured by using CPMG method, $[90^\circ_x - (\tau - 180^\circ_y - \tau)_m - \text{acq}]$, performing 34 experiments with varying the numbers of echoes.

RESULTS AND DISCUSSION

The Ternary Phase Diagram. In the following we describe the results of our investigation. The construction of phase diagram is essentials for the study of microemulsions because the hydrophilic part of the nonionic TX-100 surfactant molecule is greater than that of the hydrophobic part and for that reason the phase behavior might be very sensitive to the nature of solvent used.⁴⁰ Herein, we have taken a hydrophobic IL, i.e., $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]$, which acts as the oil phase and one of the most common nonionic surfactant, TX-100 for constructing the ternary phase diagram with water. The phase diagram of the ternary system containing H_2O , $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]$ and TX-100 was constructed through the solubility measurements and is depicted in Figure 1. The ternary phase diagram constitutes of one multiphase zone and one single phase zone. Our study was mainly focused on the single phase zone. The single phase-zone constitutes of three microheterogeneous regions.

The W/IL system, in which water is the dispersed phase and IL as the dispersion medium, IL/W, where IL is the dispersed phase and water is the dispersion medium, and the bicontinuous zone,⁴¹ where both the water and IL are present as the continuous phase are shown. These microheterogeneous regions have been detected through the measurement of bulk viscosity⁴² and the measurements of diffusion coefficients of an

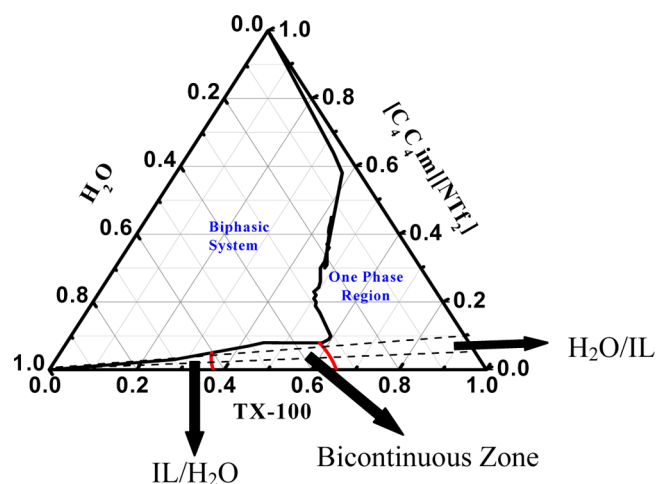


Figure 1. Ternary phase diagram of the $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ system showing change in the microheterogeneous regions from $\text{H}_2\text{O}/\text{IL}$ to bicontinuous to $\text{IL}/\text{H}_2\text{O}$ at 298.15 K. The dashed lines are chosen for studying the apparent diffusion coefficient with initial $[\text{IL}]/[\text{TX-100}]$ weight fractions are 0.05 and 0.1.

electrochemically active probe with the help of cyclic voltammetric technique.⁴³

Viscosity. To detect the microheterogeneous regions we measured the viscosity by taking $I = 0.05$ ($I = [\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]$ -to-TX-100 weight ratio). The plot of viscosity of the mixtures with the increase in the weight percent of water is shown in Figure 2. The plot shows an initial shoot in the viscosity at

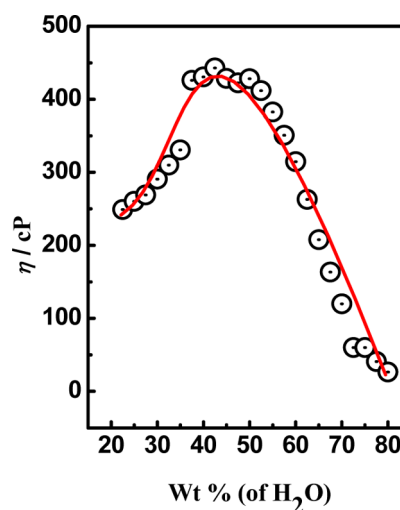


Figure 2. Dynamic viscosity, η , of the $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ microemulsions with increasing weight percent of water ($I = 0.05$).

about 40 wt % reaching to a maximum value at about 45 wt % before falling further in higher weight percents of water. There is a change in the shape of the plot at about 65–70 wt %. The initial increase in the viscosity with the addition of water suggests the formation of microstructure inside the single phase zone of the ternary phase diagram.⁴²

If there are no microheterogeneous regions inside the single phase zone, the initial addition of water should lower the viscosity of the ternary mixture. The zone up to about 40 wt % of water indicates the structural transition from W/IL microemulsions to bicontinuous structural zone supported by the reports about the structural transition from droplet W/O to

Table 1. Values of the Apparent Diffusion Coefficients (D_{app}) with the Increasing Wt% of H_2O for $I = 0.05$ and 0.1 , Respectively^a

$I = 0.05$		$I = 0.1$	
wt % (H_2O)	$D_{\text{app}} \times 10^{10}/\text{m}^2 \text{ s}^{-1}$	Wt% (H_2O)	$D_{\text{app}} \times 10^{10}/\text{m}^2 \text{ s}^{-1}$
22.5	7.1	20.0	6.6
25.0	9.8	22.5	8.0
27.5	14.8	25.0	11.9
30.0	18.6	27.5	15.6
32.5	21.0	30.0	17.2
35.0	29.2	32.5	26.9
37.5	35.3	35.0	32.1
40.0	42.1	37.5	42.1
42.5	55.4	40.0	64.0
45.0	76.6	42.5	60.2
47.5	82.9	45.0	68.0
50.0	102.1	47.5	91.7
52.5	108.5	50.0	102.1
55.0	116.4	52.5	104.5
57.5	140.4	55.0	122.0
60.0	148.7	57.5	146.4
62.5	155.2	60.0	150.5
65.0	167.7	62.5	157.6
67.5	177.2	65.0	161.2
70.0	179.5	67.5	166.7
72.5	182.0	70.0	171.6
75.0	186.7	72.5	173.8
77.5	183.0	75.0	172.3
80.0	184.3	77.5	175.2
		80.0	177.8

^aStandard uncertainties in the measurements, $u(D_{\text{app}}) = \pm 2.0\%$

bicontinuous microemulsions.⁴⁴ The deviation from the linearity observed at around 70 wt % of water is the structural transition from bicontinuous to IL/W microemulsion zone.⁴² Inside the bicontinuous microemulsions regions, the change in the viscosity behavior is may be due to the possibility of formation of some kind of structural changes at around 45 wt % of water.⁴⁴ The continuous decrease in viscosity observed in the case of bicontinuous microemulsions zones may be due to the dynamic nature of the region. In this region, the discrete particle nature of W/IL microemulsion droplets vanishes and both the water and ionic liquid phases are in a state of continuous dynamics with each other. Therefore, the viscosity of the bicontinuous microstructure decreases continuously with the increase in the weight percent of water. Again, as viscosity is a bulk property, therefore it might not give us the correct information regarding the determination of structural transition points inside the single phase zone of microemulsions. For this reason, we carried out the diffusion coefficients measurements of an electrochemically active probe inside the microemulsion system with the help of cyclic voltammetric technique. This is discussed below.

Diffusion Behavior of Electrochemically Active $\text{K}_4\text{Fe}(\text{CN})_6$ Probe Inside the Microemulsion. Different techniques like pulsed-gradient spin echo NMR,^{45–47} dynamic light scattering^{48,49} and cyclic voltammetry^{50,51} are used for the study of microstructural transition of the micellar systems. In this investigation, we used the cyclic voltammetric technique for the detection of microheterogeneous regions inside the ternary phase diagram. The apparent diffusion coefficient measurements were performed through the cyclic voltammetric technique by taking the potassium ferrocyanide [$\text{K}_4\text{Fe}(\text{CN})_6$]

as the electrochemically active probe¹⁷ inside the [$\text{C}_4\text{C}_4\text{im}$]-[NTf₂]/ H_2O /TX-100 ternary system and is represented in Table 1. The peak currents of the redox reversible systems ($\text{Fe}^{2+}/\text{Fe}^{3+}$) are measured with different scan rate and the apparent diffusion coefficients are calculated by applying the Randles-Sevcik equation given by

$$i_p = \frac{0.447F^{3/2}An^{3/2}D^{1/2}C\nu^{1/2}}{R^{1/2}T^{1/2}} \quad (2)$$

where n is the number of electrons involved in the redox reaction, A is the working area of the electrode, D is the diffusion coefficient of the electrochemically active probe in the solution, C is the concentration of the electroactive probe in the solution, ν is the sweep rate or scan rate used, F is the Faraday constant, R is the gas constant, and T is the absolute temperature.

For the study of microstructural transition it is essential that the electrochemical charge transfer must be diffusion controlled in nature.⁴⁹ The plots of peak current i_p versus $\nu^{1/2}$ shown in Figure 3 are straight line passing through the origin indicating that the electron transport reaction for the redox system inside the microemulsions are diffusion controlled in nature.^{50,51} The change in the apparent diffusion coefficient values with an increase in the weight percent of water gives us the indication about the microstructural transition points inside the single phase zone of the ternary phase diagram. For the detection of microheterogeneous regions the two I values ($I = [\text{C}_4\text{C}_4\text{im}]\text{-}[\text{NTf}_2]\text{-to-TX-100}$ weight ratio) are chosen as 0.05 and 0.10, as shown in Figure 1. Parts a and b of Figure 4 represent the plot of the apparent diffusion coefficients, D_{app} of the electrochemi-

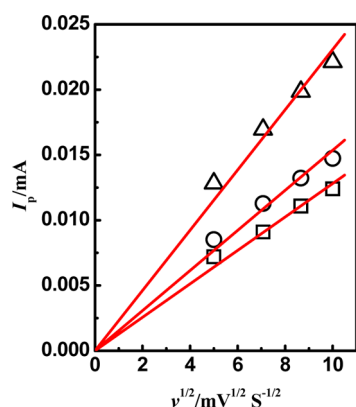


Figure 3. Plot of the peak current (i_p) versus square root of scan rate ($v^{1/2}$) for 25 wt % H_2O ($\square$), 30 wt % H_2O ($\circ$) and 40 wt % H_2O ($\triangle$) with IL-to-TX-100 weight fraction as 0.1. The lines are of the linear fit.

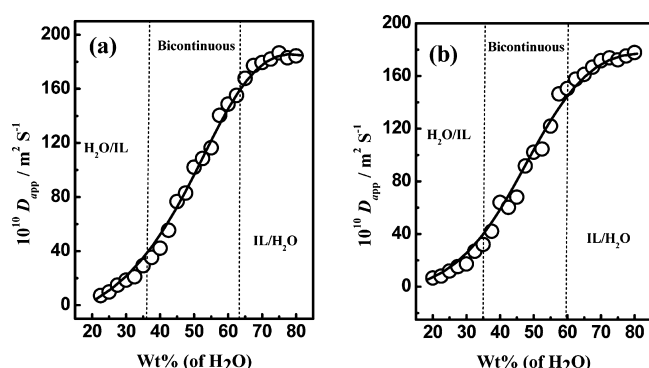


Figure 4. Apparent diffusion coefficient, D_{app} of $\text{K}_4\text{Fe}(\text{CN})_6$ in $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ microemulsion system showing the structural transition from W/IL to bicontinuous to IL/W as a function of water content at a temperature of 298.15 K. ($[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{TX-100}$ weight ratio = 0.05 (a) and 0.1 (b)). The lines are drawn to guide reader's eye.

cally active probe, $\text{K}_4\text{Fe}(\text{CN})_6$ with the increasing weight percent of water at $I = 0.05$ and 0.1 , respectively. It has been clearly shown in the above figures that with the increase in the weight percentage of water the diffusion coefficients of the electrochemically active probe are going on increases. The probe being ionic in nature is only soluble in the water phase. Initially, a mild increment in the diffusion coefficient values are observed in low weight percent of water that is below 34% for $I = 0.05$ and below 31% for $I = 0.1$. This is due to the reason that the ionic electrochemically active probe remains inside the water pool possessing high viscosity. With the increase in the weight percent of water the microviscosity of the confined water droplets are going on decreases.⁵² The D_{app} values of the probe inside the pool increases with an increase in the fluidity of droplets. A mild enhancement in the D_{app} values of the probe is due to an increase in the content of water and not due to any change in the microstructure of the microemulsions. The microemulsions formed in these regions are of the kind of W/IL type.¹⁶

The increase in the D_{app} values of the probe are again slow down in the water-rich phase zone and a mild enhancement in the D_{app} values are observed with the higher weight percent of water i.e. above 60% for $I = 0.05$ and 57.5% for $I = 0.10$, respectively. In these regions the microemulsions are of the kind of IL/W type.¹⁶ In this region the IL molecules are

trapped by the hydrophobic core of the surfactant molecules and remain as in the pool form and forms the IL/W microemulsions.¹⁶

The IL molecules act as the dispersed phase with water as the dispersion medium. The electrochemically active $\text{K}_4\text{Fe}(\text{CN})_6$ probe molecules are now in the dispersion water medium, present in the bulk quantity and the D_{app} values of probe molecules are higher and resembles to that of diffusion in the neat water.¹⁶ Thus, the increase in the amount of water has a very minimal effect on the diffusion coefficient of $\text{K}_4\text{Fe}(\text{CN})_6$ and the structure of the microemulsion is insensitive to the compositions of water in this region. However, a dramatic change in the D_{app} values is observed in the intermediate water compositions. This indicates that the environment of the microemulsion is different from that of the droplet structure.⁵³ Our results infer that when the water compositions are between 34 and 60% ($I = 0.05$) and 31–57.5% ($I = 0.1$), a bicontinuous microstructure is formed, in which both the water and hydrophobic $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]$ are present as the continuous phase.⁵³ The diffusion of molecules in the continuous medium is faster as compared to the molecules confined in the closed domains⁵³ and thus the probe inside the bicontinuous structure has higher diffusion coefficient value as compared to that of W/IL droplet microemulsion.

The increment in the diffusion coefficient values with the increase in the weight percent of water follows the sigmoidal trend and the three different kinds of microheterogeneous regions can be detected by the transition points obtained from Figure 4 are given in Table 2. Thus, from all of the above

Table 2. Microstructural Transition Points for $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ Microemulsions System at Different I Values^a

I	W/IL-to-bicontinuous transition point	bicontinuous-to-IL/W transition point
0.05	36	63.5
0.1	35	60.0

^aThe values are given in weight percent of water. ($I = [\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]\text{-to-TX-100}$ Weight Ratio).

investigations, we can deduce that three different microheterogeneous regions are present inside the single phase zone and those are W/IL, bicontinuous and IL/W microemulsions and are represented in Figure 1.

The interfacial reaction occurring on the electrode surface of both the W/IL and IL/W microemulsions are different from that of the pure water system. It has been reported that the organic species like TX-100 and $[\text{C}_4\text{mim}][\text{PF}_6]$ are active toward the surface of the electrode.⁵⁴ Being organic in nature both the $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]$ and TX-100 are active toward the surface and adsorb on the electrode surface. In the case of W/IL microemulsions, the $\text{K}_4\text{Fe}(\text{CN})_6$ molecule remain inside the water pool. Because of thermal motion, there is an exchange of masses between the nanodroplets of W/IL microemulsions containing $\text{K}_4\text{Fe}(\text{CN})_6$. Because of this thermal motion, the electrochemically active probe molecule pass through the organic surfactant layer and come in contact with the electrode surface leading to the electrochemical reaction occurs.⁵⁴ On the other hand, the electrochemical reaction for IL/W microemulsions is very similar to that of the pure water as the probe molecule reside in the water bulk phase.

Dynamic Light Scattering (DLS) Measurements. With the phase diagram and cyclic voltammetric studies yielding information on the presence of three different regions of microemulsion, we now move to investigate the system by using DLS studies to know whether the water molecules are entrapped by the surfactant molecules to form W/IL microemulsions or the IL molecules confined inside the surfactant layer to form the IL/W microemulsions. The hydrodynamic radii (D_h) of IL/W microemulsions because of their low viscosity values are given in Table 3. The D_h of

Table 3. Measured Hydrodynamic Radii (D_h) of Microemulsions Obtained Through the Dynamic Light Scattering Measurements (I and R Represent the $[C_4C_{10}im][NTf_2]$ -to-TX-100 Weight and Molar Ratios, Respectively)^a

H ₂ O/wt %	IL/wt %	TX-100/wt %	I	R	D_h /nm
67.5	1.55	30.96	0.05	0.07	4.6
72.5	1.31	26.20	0.05	0.07	5.0
77.5	1.07	21.42	0.05	0.07	5.5
67.5	2.41	30.08	0.08	0.11	5.2
72.5	2.04	25.46	0.08	0.11	5.4
77.5	1.67	20.83	0.08	0.11	5.7
67.5	2.95	29.54	0.1	0.14	5.6
72.5	2.50	25.00	0.1	0.14	5.6
77.5	2.04	20.45	0.1	0.14	5.7

^aStandard uncertainties in the measurements, $u(D_h) = \pm 0.5\%$

different IL/W microemulsions are measured by varying I values. With the increase in the I values, the D_h values of the microemulsion increase for a fixed weight percent of water (Figure 5). The drawing represents the change in the D_h values

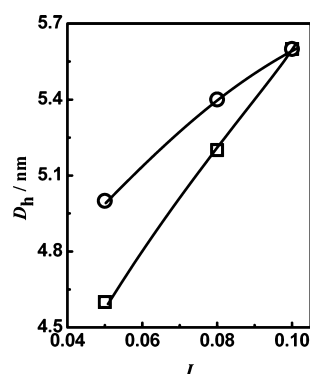


Figure 5. Hydrodynamic radii (D_h) of the IL/W microemulsion droplets with I values for $[C_4C_{10}im][NTf_2]/H_2O/TX-100$ system with a fixed weight percent of water (67.5%, □, and 72.5%, ○) showing the swelling of the microemulsion droplets.

of the IL/W microemulsion droplets with the increase in the I values for two different fixed compositions of water (67.5% and 72% of water). This is due to the reason that with the increase in I values, the content of IL in the system increases causing the swelling of the microemulsions and thereby increasing the sizes of microemulsion droplets.

Our findings are supported by the work of Falcone et al.⁵⁵ who observed that with the increase in the IL content, an increase in the droplet size of the nonaqueous microemulsions was observed. This swelling behavior of confined media is reported for aqueous microemulsion systems.⁵⁶ This suggests

that the IL molecules are entrapped by the surfactant molecules in order to form IL/W microemulsions. The increase in the droplet size of the microemulsions can also be explained with the help of packing parameter (P) and can be represented as $P = v/al$, where v , a and l signify the volume of the hydrocarbon chain, effective headgroup area of the surfactant and the hydrocarbon chain length of the surfactant, respectively.⁵⁵ It has been reported that the size of the confined media depends upon the effective packing parameter of the surfactants used.⁵⁵

Decreasing the packing parameter of surfactant molecules increases the size of confined media and vice versa.⁵⁷ With the increase in the content of IL into the system by increasing the I values, the IL molecules penetrate into the surfactant layer and interact with the headgroup of the TX-100 surfactant molecules. This interaction increase the effective headgroup area, a of the surfactant molecules. The interactions between the cationic part of $[C_4mim][PF_6]$ with the polar headgroup of nonionic TX-100 surfactant in the case of $[C_4mim][PF_6]/H_2O/TX-100$ microemulsion system also supports the above statement.¹⁴ Thus, the value of a increase which in turn decrease the packing parameter, P of the surfactant molecules. Consequently, the droplet size of the IL/W microemulsions increases. It is suggested in the literature that a linear dependence of the D_h values on I points to spherical droplets of microemulsions.⁵⁸ However, a slightly nonlinear dependence in our case seems to be with the increase in the weight percent of water to 72.5%. However, we currently are not aware of any reasons behind it.

¹H NMR and NMR Relaxation Study. Many workers have studied the properties of this water present inside the reverse micelles through the help of different experimental techniques like DLS,^{48,49} IR spectroscopy,⁶ UV-vis spectroscopy,^{59,60} fluorescence spectroscopy^{40,52} and NMR spectroscopy.^{3,4} In most of these reports different spectroscopic probe molecules are used for the determination of physicochemical properties of different confined media. The location of probe molecules inside the reverse micelles however creates problems in finding the true physical properties of these confined media. In our work, we have carried out the NMR spectroscopic measurements of water inside the W/IL confined media, which gives the information related to the physical properties of these confined media without the help of any spectroscopic probe.^{61,62} Figure 6 represents the stacking of ¹H NMR spectra of different W/IL microemulsions with the increase in the weight percent of water. The upper and lower parts of the drawing illustrate about the IL-to-TX-100 weight ratio (I) as 0.1 and 0.05, respectively. It has been shown in Figure 6 that with the increase in the water content from 5.0 wt % to 27.5 wt %, the peak at around 4.0 ppm (5.0 wt % of water) is nearly shifted to 0.4 ppm toward the deshielded region leaving rest of the peak unchanged. It signifies that the peak is the indication of the proton of water molecules present inside the reverse micelles. To confirm the water signal emerging out from the microemulsion samples, we also carried out the two-dimensional ¹H–¹³C heteronuclear single quantum coherence spectroscopy (HSQC) and the results are shown in Figure 7.

The drawing represents the HSQC spectra for 22.5 wt % of water with IL-to-TX-100 wt. ratio, $I = 0.05$. As we have recorded the ¹H–¹³C HSQC spectra the water signal must not have the contour plot due to the absence of carbon atom in the molecule. The enlarged bottom view of the graph shows the absence of such contour plot at 4.27 ppm on the ¹H NMR

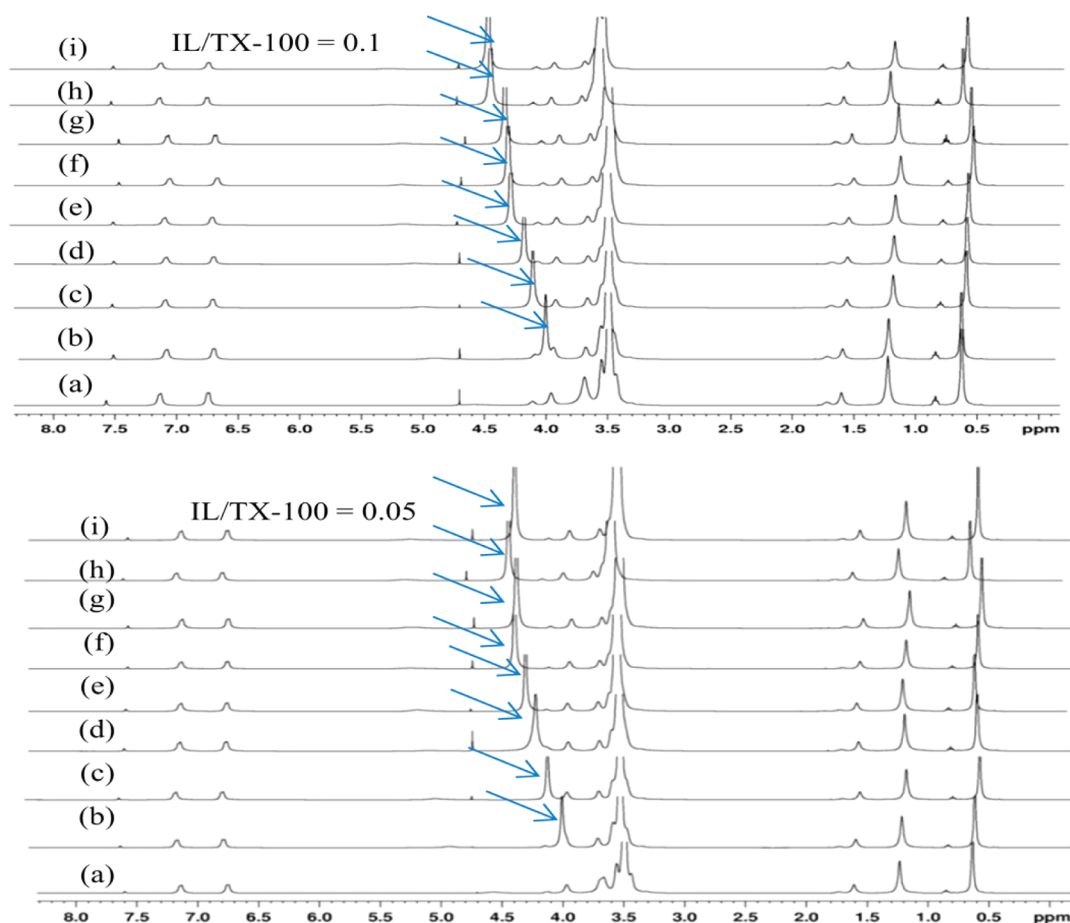


Figure 6. Stacking of ^1H NMR spectra showing the types of H_2O molecules present inside the W/IL microemulsions with different weight percent of water: 5.0 wt % (a), 7.5 wt % (b), 10.0 wt % (c), 15.0 wt % (d), 17.5 wt % (e), 20.0 wt % (f), 22.5 wt % (g), 25.0 wt % (h), 27.5 wt % (i). Top: weight ratio of IL/TX-100 = 0.1. Bottom: IL/TX-100 = 0.05).

spectra axis, which confirms the signal emerging from the water molecule.

With the lower weight percent of water, the peak coming out from the water molecules are mixed up with the proton of the TX-100 molecules.⁶³ H-bonding interaction between the $-\text{OH}$ group of TX-100 molecules with the water molecules inside the $[\text{C}_4\text{mim}][\text{PF}_6]/\text{H}_2\text{O}/\text{TX-100}$ microemulsions supports the above statement.¹⁴ This suggests that the interfacial water present around the headgroup of TX-100 molecules experience a different interacting environment as compared to that of the core water molecules. It was reported that there are three types of micellar water are present; namely trapped water, bound water and free water.⁶⁴ The dissolved water molecules present in the nonpolar solvents are known as trapped water molecules. Water molecules, which are H-bonded with the headgroup of surfactant molecules, are called as the bound water molecules and the core water molecules of the reverse micelles are termed as the free water molecules. The initial addition of water to the $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ microemulsions system binds with the headgroup of surfactant molecules through the help of hydrogen bonding interaction between the H atom of water with the oxygen atom of oxyethylene unit of TX-100 and experiences a shielding effect.⁶⁵ With the increase in the weight percent of water, the water molecules form a pool inside the surfactant layer and behave like that of the bulk water molecules. These water molecules do not interact with the headgroup of surfactant molecules and are termed as free water

molecules with a slightly deshielding effect as compared to that of the bound water molecules. It means that the protons of bound water molecules are experiencing a higher electron density as compared to that of the protons of free water molecules. The lower IR stretching frequency of $-\text{OH}$ bond of free water molecules ($3220 \pm 20 \text{ cm}^{-1}$) as compared to that of bound water molecules ($3400 \pm 20 \text{ cm}^{-1}$) also supports the above statement.⁶⁵

^1H NMR spin–lattice relaxation time (T_1) study of water molecules was carried out to understand the dynamics of the water molecules present inside the $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ microemulsions. By applying the Bloembergen, Purcell, and Pound (BPP) theory the reorientational correlation time (τ_c) of water molecules can be calculated from the measured T_1 relaxation time of water molecules and are illustrated in Figure 8.⁶⁶ According to BPP theory both the spin–lattice relaxation time, T_1 and reorientational correlation time, τ_c are inversely related to each other as represented in the following eq 3.

$$\frac{1}{T_1} = \frac{3}{10} \frac{\gamma^4 \hbar^2}{b^6} \left[\frac{\tau_c}{1 + \omega^2 \tau_c^2} + \frac{4\tau_c}{1 + 4\omega^2 \tau_c^2} \right] \quad (3)$$

where T_1 is the NMR spin–lattice relaxation time, γ the gyromagnetic ratio of the proton, $\hbar$ the Planck's constant, b the intraprotion distance of water molecule, τ_c the reorientational correlation time and ω the resonance frequency. Parts a and b of Figure 8 represent the variation of T_1 and τ_c values,

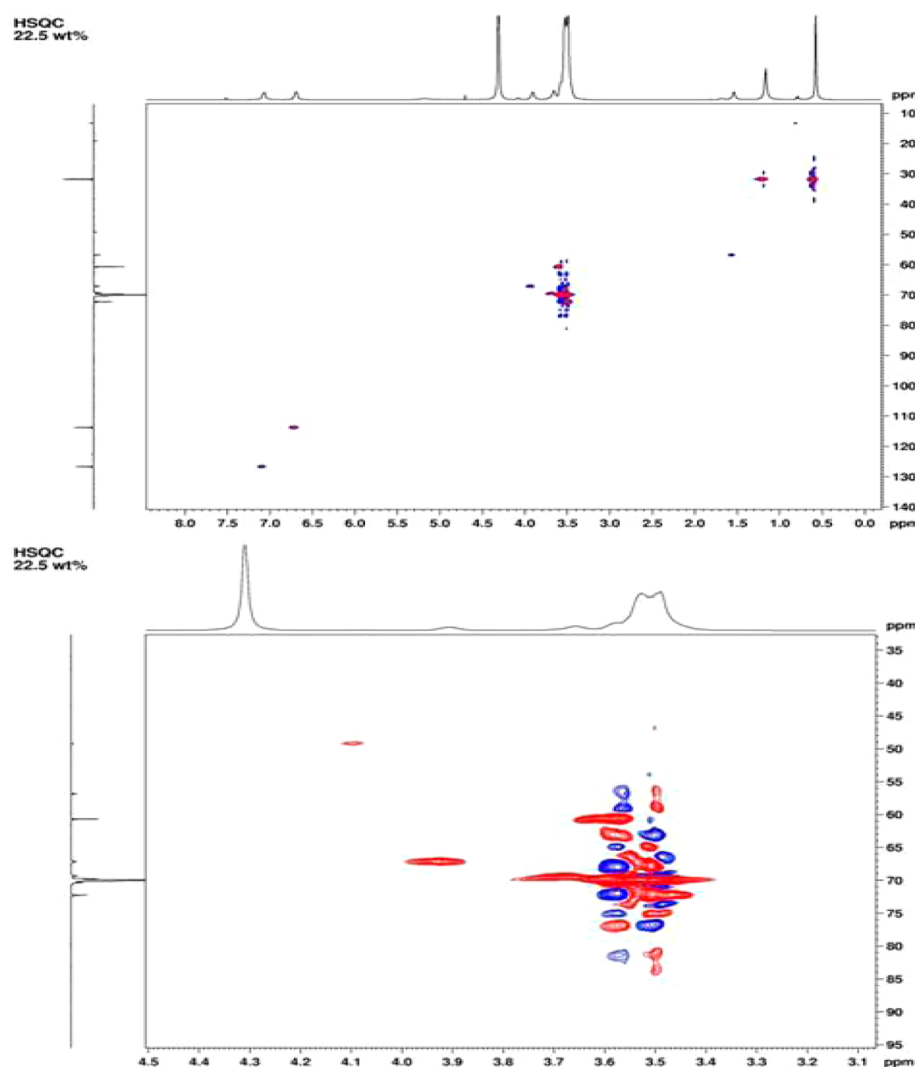


Figure 7. NMR ^1H – ^{13}C heteronuclear single quantum coherence (HSQC) spectra of W/IL microemulsion containing 22.5 wt % of water showing the confirmation of H_2O signal at 4.27 ppm. ($I = 0.05$).

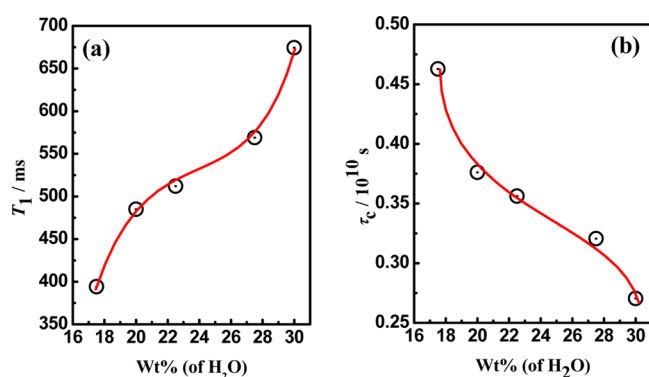


Figure 8. Variation of NMR spin–lattice relaxation time T_1 (a) and their calculated reorientational correlation time τ_c (b) with the increase in the weight percent of water.

respectively, with the increase in weight percent of water. It is seen from the plots given in Figure 8 that with the increase in weight percent of water, the T_1 values are increase whereas the τ_c values decrease. It is reported that the relaxation time observed in the case of H_2O /hexanol/TX-100/cyclohexane was the weighted average of the signals coming out from the proton

of the H_2O molecules, $-\text{OH}$ group of hexanol and the $-\text{OH}$ group of the TX-100 molecule.⁶¹ As in the case of $[\text{C}_4\text{C}_{4}\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ microemulsion systems, the TX-100 molecule contain a hydroxyl proton, there is the possibility of fast exchange of proton takes place between this proton with the proton of water molecules which must be faster than our observed relaxation time. Therefore, the observed T_1 values are the weighted average of these two signals. The increase in the T_1 values with the increase in the weight percent of water indicates that the mobility of proton inside the micellar core increases as confirmed by the decreased reorientational correlation time, τ_c shown in Figure 8b. The τ_c values of water molecules present inside the micellar core of the $[\text{C}_4\text{C}_{4}\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ microemulsions are in the range of 0.25×10^{-10} to 0.46×10^{-10} s. It is reported that the overall tumbling motion of W/O microemulsions are of the order of 10^{-9} s.⁶⁷ If we consider the tumbling motion of these W/IL microemulsions in the range to that of the W/O microemulsions, we can then signify that the motion of water molecules is not coupled with the overall tumbling motion of these W/IL microemulsions and is different from them.

Therefore, the bulk transport property like viscosity of these W/IL microemulsions is different from that of the microscopic

transport property like microviscosity of the confined water. Reports are available in the literature regarding the determination of microviscosity of the reverse micelles or microemulsions by the use of different spectroscopic probe molecules.^{2,52} The location of probe molecules inside the micellar core are however questioned by many workers leading to a problem in the determination of exact microviscosity of micellar core.^{3,4} In our study, we attempted to estimate the η_{micro} of the micellar core of the $[\text{C}_4\text{C}_{4}\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ microemulsions system with the help of NMR spin–lattice relaxation (T_1) measurements.⁶² Assuming the Debye-Stokes equation to be valid in these microemulsion systems the microviscosity of the micellar core is estimated from the reorientational correlation time as

$$\tau_c = \frac{4\pi na^3}{3kT} \quad (4)$$

where a is the radius of the molecule, k is the Boltzmann constant, T is the absolute temperature and η is the microviscosity experienced by the molecule inside the micellar system.

From Figure 9, it is seen that the η_{micro} experienced by the water molecules inside the water pool of W/IL microemulsion

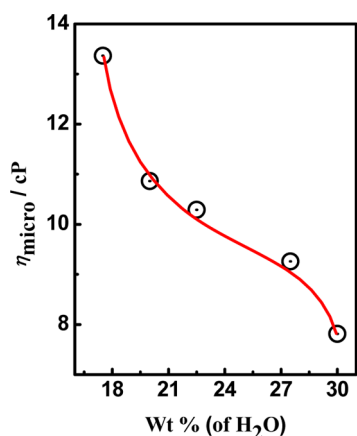


Figure 9. Variation of the microviscosity of the water present inside the micellar core of the $[\text{C}_4\text{C}_{4}\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ system with the increase in the weight percent of water for IL-to-TX-100 weight fraction, I as 0.05.

decreases with increasing weight percent of water. The physicochemical properties of aqueous confined media of small reverse micelles are significantly different from that of the bulk water,^{7,52} whereas with the increase in the size of the aqueous micellar phase, its properties approaches to that of bulk liquid water.^{61,62} With the lower content of water, it interacts with the nonionic headgroup of TX-100 molecules.¹⁴ Because of this the water molecules present near to the surfactant headgroup experienced a higher viscosity and these water molecules are termed as the bound water molecules.⁶⁴ The higher correlation time value for the lower weight percent of water also supports the above statement. With the higher water content, the water molecules lose their connectivity with the headgroup of surfactant molecules and form pool of water molecules inside the micellar core.^{61,62} Therefore, these water molecules experience a higher fluidity and in turn a decrease in their microviscosity values.^{61,62} These water molecules are termed as free water molecules, and their NMR reorientational

correlation time remains invariant with the addition of water.⁶⁷ We have however observed a continuous decrease in the correlation time of the water molecules and their microviscosity values with the increase in the weight percent of water.

This contradictory observation may be due to the reason that the so-called free water molecules present inside the micellar core of the microemulsions are not actually free from the interacting influence of the headgroups of TX-100 surfactant molecules. Another interesting observation in the case of these W/IL microemulsions is that there exist an inverse relationship between the microviscosity and the bulk viscosity as illustrated in Table 4.

Table 4. Comparison Between the Microviscosity (η_{micro}) with the Bulk Viscosity (η_{bulk}) and the Spin–Lattice Relaxation Time (T_1) with the Relaxation Rate (R) of the Microemulsion Samples with the Increase in the Weight Percent of Water ($I = 0.05$)^a

SL no.	wt % of H ₂ O	η_{micro} /cP	η_{bulk} /cP	T_1 /ms	$R = (1/T_1)/\text{s}^{-1}$
1	17.5	13.36		394.20	2.54
2	20.0	10.86		484.99	2.06
3	22.5	10.29	248.90	512.06	1.95
4	27.5	9.26	269.10	569.08	1.76
5	30.0	7.81	290.80	674.50	1.48

^aStandard uncertainties in the measurements, $u(\eta_{\text{bulk}}) = \pm 2.5\%$, $u(T_1) = \pm 0.1\%$

It means, within the studied W/IL microemulsion zone, that the increase in the amount of water decreases the microviscosity of the water nanodroplets; on the other hand it increase the bulk viscosity of the microemulsion mixtures. The decrease water–surfactant interaction is the primary cause for the decreased η_{micro} , however the increase droplet–droplet interactions with the increase in the weight percent of water are responsible for the enhancement in the η_{bulk} of the W/IL microemulsion system. Also, the relaxation rate of the water molecules in the W/IL microemulsion zone decrease with the increase composition of water. It means that the rate of relaxation of water molecules is directly related to the microviscosity of the nanodroplets. Higher is the microviscosity of the medium, higher is the time required for the proton of water molecule to be relaxed. As both the free and bound water molecules are present inside the W/IL microemulsion systems, therefore it is of interest to know the fraction of each water molecules. For this purpose, we used the model of Hansen for W/O microemulsions system.⁶² According to him

$$T_1^{-1} = f_B T_{1B}^{-1} + f_F T_{1F}^{-1} \quad (5)$$

where T_1 , T_{1B} , and T_{1F} are the experimentally observed relaxation time, relaxation time of bound water molecules and relaxation time of free water molecules present inside the micellar core, respectively. f_B and f_F are the mole fraction of bound and free water molecules, respectively and $f_B + f_F = 1$.

By replacing the relaxation rate, R in place of relaxation time, T_1 , eq 5 can be rewritten as

$$R = f_B R_B + f_F R_F \quad (6)$$

where R , R_B , and R_F are the observed relaxation rate, relaxation rate of bound water molecules and relaxation rate of free water molecules, respectively. Rearranging eq 6, the fraction of bound

water molecules can be calculated by applying the following eq 7.

$$R = f_B R_B + (1 - f_B) R_F$$

$$f_B = (R - R_F) / (R_B - R_F) \quad (7)$$

Assuming the relaxation rate of free water molecules is similar to that of the bulk water molecules as 0.278 s^{-168} and that of bound water molecules as 14.08 s^{-1} , which is obtained from the intercept of the linear fitting of relaxation time with the weight percent of water. Figure 10 represents the calculated

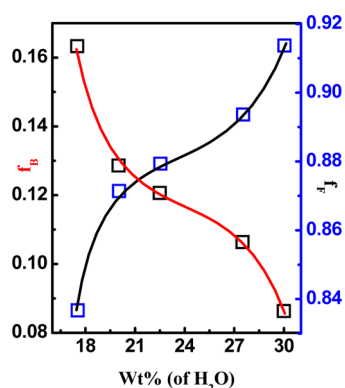


Figure 10. Variation of mole fraction of bound water molecules f_B (red line) and free water molecules f_F (black line) which are present inside the aqueous nanodroplets of $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ microemulsion systems with the weight percent of water.

mole fraction of bound and free water molecules with the increasing composition of water. The mole fraction of bound water molecules is decrease where as the mole fraction of free water molecules is increase with the increase in the composition of water. This suggests that the probability of interactions between the water molecules with the surfactant molecules is increase with the decrease composition of water where as increasing water content strengthen the water–water interactions.

It also infers that the reverse micelles contain higher amount of free water molecules as compared to that of the bound water molecules within the studied W/IL microemulsions zone. Reports are available in the literature that the mole fraction of bound water gets saturated beyond a particular composition of water.⁶² However, in the case of W/IL microemulsions zone of $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ microemulsions we have observed a continuous decrease in the value of mole fraction of bound water molecules as illustrated in Table 5 and is confirm from the observed spin–lattice relaxation time T_1 .

Reports are available in the literature regarding the determination of pH of the aqueous micellar media by the use of different spectroscopic molecular probe.^{3,69} However, as

Table 5. Mole Fraction of Bound Water Molecules (f_B) with the Increase in the Weight Percent of Water

wt % (H_2O)	f_B
17.5	0.16
20.0	0.13
22.5	0.12
27.5	0.11
30.0	0.09

stated above, the use of probe molecules creates problem in finding out the exact physical properties of different micellar media because of its difficulty in find out the exact location of it inside the reverse micellar media.^{3,4} To avoid this problem, we have used the NMR spin–spin relaxation time, T_2 of water molecules as a probe to measure the pH of the aqueous reverse micellar core of the $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ microemulsion systems, as T_2 relaxation time is sensitive to the pH of the aqueous micellar medium.⁶³ Figure 11 represents the

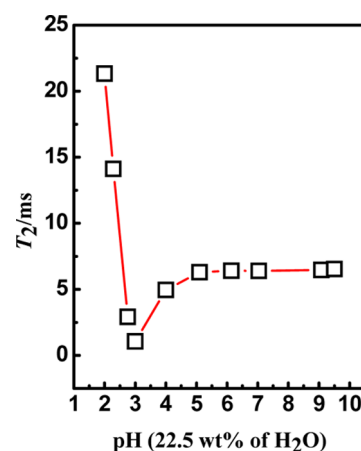


Figure 11. Variation of NMR spin–spin relaxation time, T_2 of water molecules (signal at 4.27 ppm) with the pH of the micellar core for the $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ microemulsion system with a fixed weight percent of water as 22.5 wt %. The pH of the aqueous micellar systems are externally measured before the use of it for the experimental purpose. ($I = 0.1$).

variation of NMR spin–spin relaxation time, T_2 of water molecules present inside the reverse micellar core of the $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ microemulsion systems with the pH of the medium. At higher proton concentrations, the NMR spin–spin relaxation time, T_2 of water molecules shows its maximum value as 21.336 ms. However, with the increase in the pH of the aqueous micellar medium this value sharply drops to a minimum at 1.06 ms as represented in Table 6. Afterward a continuous increase in the T_2 relaxation time values were observed with the increase in the pH of the aqueous reverse micelles.

The reason for this variation in T_2 relaxation time values are due to the possibility of variation in the rate of proton exchange

Table 6. Influence of pH on the Spin–Spin Relaxation Time, with T_2 of Water Molecules Present Inside the Micellar Core of $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ Microemulsions with IL-to-TX-100 Weight Fraction as 0.1^a

pH	T_2/ms	$\Delta\nu_{1/2} (1/\pi T_2)/\text{ms}^{-1}$
2.00	21.336	0.015
3.00	1.060	0.300
4.00	4.950	0.064
5.10	6.302	0.051
6.14	6.408	0.050
7.03	6.403	0.050
9.08	6.474	0.049
9.50	6.546	0.049

^aThe line widths of the spectra are calculated from the measured T_2 values. Standard uncertainties in the measurements, $u(T_2) = \pm 0.1\%$.

between the water molecules and the hydroxyl proton of TX-100 molecules.⁶³ At the high proton concentration, the rate of proton exchange between the water and TX-100 molecules are faster. Here, the magnetically excited protons switch their precessional frequency faster causing slower dephasing of their transverse magnetization vector and consequently, we obtain a higher value of T_2 relaxation time.⁷⁰ The calculated line width from the measured T_2 relaxation time given in Table 6 also supports the above statement. At the higher proton concentration and with the maximum value of T_2 relaxation time the calculated line width of the signal is minimum as 0.015 ms^{-1} which indicates the sharpening of the signal that emerges out from the protons of both the water and TX-100 (alcoholic proton) molecules due to the faster rate of exchange between them.⁷¹ At the minimum value of T_2 relaxation time (1.06 ms at pH = 3) the rate of proton exchange is in between the fast and slow intermediate regimes.⁷⁰ Here the calculated line width of the signal is highest as 0.3 ms^{-1} which indicates the broadening of the peak due to the fusion of the two peaks coming out from the water and TX-100 molecules. The peak coalesces also occurs at the minimum value of T_2 relaxation time between the protons of water and $-\text{OH}$ group of hexanol molecules inside the CTAB microemulsions.⁶³ However, with the increasing the pH of the aqueous reverse micellar media of $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ microemulsion systems from 3.0 to onward, the concentration of proton inside the aqueous confined nanodroplets decrease and consequently the rate of proton exchange decreases. The decrease in the proton concentration creates a heterogeneous environment in the local magnetic field inside the aqueous IL microemulsion systems causing the faster dephasing of the transverse magnetization vector.⁷⁰ This results in the decrease in the values of the T_2 relaxation time with the increase in the pH of the aqueous IL micellar core. The calculated line width of the water signal is around 0.05 ms^{-1} and remains invariant with the further increase in the pH of the aqueous IL microemulsions. This may be due to the splitting of the two proton signals coming out from the water and TX-100 molecule (alcoholic proton).

We also investigate the micropolarity of the aqueous nanodroplets of $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ microemulsions by the use of methyl orange (MO) as the UV-vis spectroscopic molecular probe.

Because of the hydrophilic nature of MO, it is soluble in the aqueous micellar core and insoluble in the hydrophobic ionic liquid phase. Therefore, MO is considered to be a very good polarity probe for the determination of micropolarity of the microheterogeneous medium. The electronic transition energy coming out from the measured λ_{max} values of MO is sensitive to the polarity of the microenvironment. The measured λ_{max} values of aqueous confined media of $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ microemulsions are given in Table 7 for the IL-to-TX-100 wt. ratio I , as 0.05 and 0.1, respectively. From the above table it is very clear that the λ_{max} values of MO increase with an increase in the weight percent of water. This indicates that the micropolarity of the aqueous confined media increases with the increase in the λ_{max} values of MO.⁴⁰ The variation of λ_{max} values of MO with the weight percent of water is illustrated in Figure 12. We have observed nearly a linear dependence of λ_{max} values with the change in the composition of water agreeing with the published report.⁷² This linear dependency of λ_{max} values of MO in case of $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ microemulsion systems indicates that the water molecules present inside the micellar core are not completely free from the interacting

Table 7. Micropolarity of Aqueous Confined Media of $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ Microemulsions with the Increasing Weight Percent of Water (IL-to-TX-100 Weight Ratios as 0.05 and 0.1, Respectively)

IL/TX-100 = 0.05		IL/TX-100 = 0.1	
wt % H_2O	λ_{max}	wt % H_2O	λ_{max}
10.0	426.0	7.5	427.9
15.0	428.0	12.5	429.0
22.5	429.0	20.0	429.9
27.5	430.1	25.0	431.0

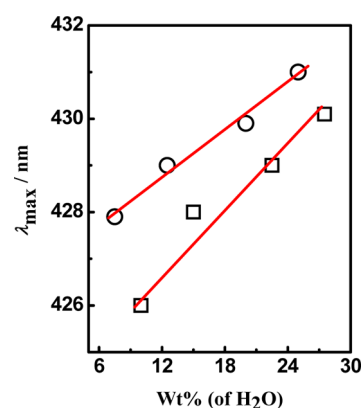


Figure 12. Variation of λ_{max} values of MO as the UV-vis spectroscopic absorption probe as a function of weight percent of water for $I = 0.05$ (□) and 0.1 (○), respectively.

influence of the polar hydrophilic part of the nonionic TX-100 surfactant molecules as discussed above in the NMR relaxation studies.

CONCLUSIONS

In summary, we show that

- (1) The NMR T_1 relaxation time of water molecules can be used as an alternative parameter to indicate the microviscosity of the $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ microemulsion systems.
- (2) Both the microviscosity and the bulk viscosity values of $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ microemulsion systems are inversely related to each other.
- (3) The NMR T_2 relaxation time of water molecules varies with pH of the $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ microemulsion systems and gives an indication of the pH of the aqueous micellar core.
- (4) Two kinds of water molecules (the bound water and free water molecules) are present inside the $[\text{C}_4\text{C}_4\text{im}][\text{NTf}_2]/\text{H}_2\text{O}/\text{TX-100}$ microemulsion systems. The fraction of bound water molecules decreases whereas the fraction of free water molecules increases with an increase in the composition of water. However, at each composition of water the fraction of free water molecules are higher as compared to that of fraction of bound water molecules.

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Notes

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